-Gravity

Concepts

1. *Everything that has mass, pulls things around it towards itself. This pull, this force is called Gravity.*
2. *Every particle, every object, every being, every celestial body, everything has mass, even if it is a teeny tiny bit of it - and they all exhibit Gravity - they all pull everything around them. I'm pulling you, you are pulling me, the Earth is pulling all of us, you are pulling the moon, the moon is pulling you. You are pulling all the tiny particles around you, you are pulling the Earth, The Earth is pulling you and all the particles around you. Gravity is everywhere. (F-G*m1*m2/r2)*
3. *Imagine, you were just floating in space and there are no planets or stars or anything large around you for billions of miles. You would just be floating in space, in the same direction & speed you started with forever. (You can move your arms, legs, head, turn your body, but you can't change your direction or speed. Can't you swim? Not really because, for that you need enough particles with enough mass around you to push you back when you push them as you move your hand.) Now imagine Earth appears say a 100 miles from you, you would swiftly be pulled towards it, because it is so huge compared to you & can pull you easily, which it does naturally. But the moment Earth appears near where you are floating, not only does the Earth pull you, you starting pulling it too. But you wont end up moving the Earth because , as a comparatively smaller object, your gravity is not enough to pull a larger body. Similarly the Sun is pulling the Earth, other planets, other celestial bodies around itself. (are you pulling all the tiny particles around you? Are they all getting stuck to you ? or are they all circling around you, like Earth is around the Sun? Or are they all falling to the ground? What do you think is happening & why?)*

4. *But, we said everything ,pulls. Does photon also have gravity? It has no mass. Its the only "particle" we know for sure has o mass at rest. - It does have gravity. Because turns out, it is not mass directly that causes gravity, but it is actually Energy. A photon that is travelling at speed C from the sun, has energy owing to its motion. This energy causes gravity. Earth also has Energy - Its energy is more in the "frozen" form so to speak, which manifests as mass. We see the mass first and think, that is causing the Energy (E=mc2) - but thats just how we measure Energy. It is infact energy that causes Mass. Mass is one form of Energy.*
5. *How does gravity arise? - Hold a blanket from 4 sides, drop a ball on it - it curves & stretches? - All particles, objects, beings, planets, and the Universe can be imagined to be suspended on a blanket like this - The Space-Time Blanket. Just as the ball, with some mass causes the blanket to curve, so does every particle, object, etc. (imagine the space, in which I am, you are, earth is, stars are, everything is - as a blanket stretched across time - there was space billions of years ago, there will be space billions of years from now, as long as Universe exists, it can only exist on space, for a time or maybe even forever, we dont know, unless we believe Universe began with Big Bang, so if there was a start, maybe there would be an end too; then another start , another end and so on.)*
6. *So an object as big as Earth, curves this Space Time blanket. On your blanket, drop a smaller ball near where you have placed the bigger ball. What happens? Does the ball roll along the curvature caused by the bigger ball, towards the center of the curvature? This curvature caused by the mass of an object is what causes this "pull" towards itself. Same applies for the massless photons too - their speed/energy/momentum interacts with space time bending it , causing Gravity.*
   So I'm not really pulling you. There is no direct effort or energy. Earth is not pulling everything around it, Sun is not pulling the Earth. We each

exist, we are here and that causes the curvature as a result, and the curvature in turn causes other objects to fall towards it. So the Gravity, we call a force, doesn't need energy at all. Try pulling your friend, are you spending energy? Do you feel tired after a while? Is it work? - The gravitational pull is unlike this full pull. Cos the planet or the object is not doing any work or expending its energy. Energy (sometimes as mass) automatically alters the curvature. We fall along the curvature. The Energy of the object does not decrease because of this. (it could end up increasing because, once we fall into the center and burn, that energy is added on to that we fell into.)

7. *Lets go back to you floating in space, Earth is now there all of a sudden, so space time fabric is curved, and you are naturally standing sliding into the curve to its center, which will be at the geometric center of our spherical earth. (we will see more about this later) . For now, you move along the curvature towards the center of the mass . But you land on the surface of the Earth , i.e. the ground. With this ends your flow along the Space time curve created by the Earth, because you can pass through the ground, you cannot reach the center of the mass/center of the ST curve. So Gravity , the space time curve is leading you to its center which is stopped by the ground and the opposing force, a push force that it exerts on you. You are able to stand on the ground, not because Gravity is pulling you towards itself, because if it can succeed, you would probably be dragged through a hole you will pierce as you move down the ground to the center of the earth, but because, the ground is pushing you and holding you. What is this push force you experience ?- the normal force - everything has a push force too - which acts when you apply a force against it , it pushes you - only upon contact, when my atoms and the objects atoms come close to each other and start repelling. Unlike gravity which is omnipresent. On solid ground the push force, is equal to the pull force of earth's gravity. So you stand and dont get*

burrowed. On water, you float if the push force is equal to the force of gravity pulling towards the center - you sink, if the push force of the water is less. That's why you cannot float when the volume of water is not enough for your weight or when in an ocean you dont align your body in a way that compensates for the push force. If you are falling into air, its push force is much lesser than the earth's gravitational force & we would fall through. This Normal force that objects, particles, materials, exhibit depends on the density of the material - how densely the atoms are packed - more densely packed, they repel the force we apply better and are more solid. It is the Normal force = Fg =mg that the ground applies on me when I'm standing. It is the normal force = F i apply with my finger on a door , that i experience & keeps my digging into the door. Normal force is always equal and opposite to the force exerted upon itself. Fn while running on ground is usually 2-3 mg*

8. *Now, lets say, Jupiter appears, instead of the Earth. It is also huge compared to you & has a great energy and mass. You will as usual get dragged along the space time curvature it creates, which is also at its center. But with Jupiter you wont land anywhere. Jupiter is a Gas giant. It has no surface, no solid ground. Just gas, which gets thicker and thicker & hotter & hotter as you go down its center. So you will at some point be obliterated as you would if you were to fall through earth's ground into its center too. Same case with the other gas giant , Saturn. But they look solid on photos? Its just the atmosphere and gas clouds we are seeing. Nothing but more gas below it. Neptune & Uranus on the other hand are ice giants - made of h20,ch4 & nh3(ammonia) - so its like a liquid swirling sphere colder on top, gets hotter towards the core. You would again not land, but sink, until the pressure & heat devours you.*
9. *Now say a black hole appears- you would again fall into it, since its extremely high Energy & mass are confined into what we imagine as a*

point, there is no landing, you would just fall towards the center and as usual be ripped off.

10. *But, if we imagine a blanket and a ball on it, the center of the space time curvature is at the bottom of the ball, where it is touching the blanket. But we are saying the center of curvature with fall at the geometric center of the object. How? - Because space is not like a blanket, its not flat, 2dimensional - space is all around you, as you can see,and experience , in all directions & all dimensions. So when a space time curves because of Energy/Mass, it is not just in one direction, like how a blanket forms a curve only in one direction , but the space time curves radially in all directions around the object , thereby the center of curvature coinciding with the center of the object. This is not easy or even really possible to imagine these curvature, like how we can imagine a blanket curve. So we use that as an analogy to understand the behaviour. It does not exactly represent how ST behaves.*
11. *Now if gravity pulls all objects around its center, then what happens with photons, that have no mass & are in motion?*
12. *When we are floating in water we don't really feel our weights, we feel almost weightless, when standing in water we feel like we weigh lesser than on ground, when floating in space, we feel nearly 0 weight - all because of the Normal Force of the ground or the water or the air - Gravity remains the same on ground or in a pool. Gravity of the earth is pulling you down with the same force - but the opposing normal force exerted by the ground is equal and opposite to mg, which is weight on ground is mg. But in water, its opposing normal force is lesser so our weight we feel is equal to that opposing force, and hence feel lighter. In air or space, we feel almost no Normal force because there is no dense object we are pressing against - even though gravity continues to act wherever we go . Although as we go away from the earth, the earth's gravitational pull on us reduces, but we would also get into the gravitational fields of other celestial objects - all these*

gravity is constantly acting upon us, that we dont feel, cos we perceive gravity only through a normal force. We might experience 0 gravity, if we are able to reach deep space with no large bodies for millions of miles around us. Just because we feel weightless in space or while floating in an ocean, doesnt mean we are experiencing 0 gravity - it just means we are experiencing near weightlessness. Even inside the ISS - people float, not because they arent experiencing the earths gravity - they are, the ISS is - the ISS and everything inside is being pulled by the g (90% of whats on ground), but it is maintaining a velocity & sideways thereby revolving aru=ound the earth (15-16 orbits per day at 7.66km/S velocity) to keep it from falling into earth . The people and objects inside ISS are floating because, normal force is 0, not because gravity is 0.

13. *ISS is surfing gravity waves - maintain a (orbital) velocity in sideways direction to the gravity - voila - surfing - while surfing too, the water's slope pulls you down, you maintain a velocity and move sideways you can stay on the slope.*
14. *We saw how gravity bends space time , cos of which objects fall towards the center - how does it affect time? Time is slower closer to the center of gravitational field - Time Dialation Story*
15. *Is Gravity a quantum force - like the electromagnetic force, or atomic force, nuclear force ? - dont know yet -Gravitons ?*
16. *Universe - dance between the expanding force of dark energy and the attractive force of gravity - galaxies hold up, stars and plants stay clustered together within an expanding universe due to gravity*
17. *Gravitational lensing - light -photons -like any other particle or object also travels along the space time curvatures - leading to the them changing directions instead of their usual straight line -*
18. *Gravity weakens inside its surface and becomes 0 at the center - gravity is strongest at the surface.*

19. *Escape Velocity - can you jump so high that you could fly into space?*
20. *Gravitational potential energy*
21. *Center of Gravity and alignment - CoG must be vertically aligned with CoB - not only must the push force - normal force on solid and buoyant force of liquid be equal, they must also be aligned.*
22. *Trampoline - normal force is greater than gravitational pull when we apply an extra downward push against the trampoline - hence we shoot upwards . If we are rest on the trampoline - then its normal force =W, if we give an additional push, then it pushes back with N>W+Fextra - which comes from the elastic force*
23. *Gravitas - A person's quiet authority, presence - from the same root word gravis = heavy that gave us gravity,grave, grievance, aggravate - **gravity** pulls on objects physically. **gravitas** pulls on people emotionally and socially.*
24. *Gravitational waves - ripples in ST due to massive accelerating bodies. This wave also propagates at the speed of light*
25. *Gravitational Redshit*
26. *Non-uniform gravity fields*
27. *Gravitational acceleration variation with latitude and altitude*
28. *Pendulums & orbits as practical applications*
29. *Loop quantum gravity / string theory perspectives*
30. *Tides (sun-moon gravity)*
31. *Lagrange point*
32. *Multi-body gravitational systems*
33. *Cavendish experiment (measuring G)*
34. *Gravitational redshift experiments*
35. *GPS time dilation corrections*
36. *LIGO & gravitational waves*
37. *Frame-dragging measurements*

38. *Schwarzschild radius*
39. *Variable g on different planets & moon*
40. *Gravity assist (slingshot effect)*
41. *Frame dragging / Kerr black holes*
42. *Gravitational anomalies (local variations)*
43. *Artificial gravity (centripetal)*

Gravity - Scientists

Period / Century	***Region / Tradition***	***Name***	***Contribution to Gravity***
c. 6th–2nd BCE	*India*	*Kaṇāda*	*Vaiśeṣika philosophy — gurutva (heaviness) as inherent property causing falling.*
c. 4th–2nd BCE	*India*	*Aryabhata*	*In Āryabhaṭīya described Earth's rotation, explained celestial motion without falling sky.*
7th century CE	*India*	*Brahmagupta*	*Used term gurutvākarṣaṇam — Earth attracts and holds things.*
5th–12th century CE	*India*	*Sūrya Siddhānta (anon.)*	*Orbital models acknowledging attractive force between Earth and heavenly bodies.*

12th century CE	*India*	*Bhāskara II*	*Expanded on gurutva and falling objects.*
Ancient	*India*	*Jyotisha astronomers*	*Empirical observation of Sun & Moon influence (tides, eclipses).*
4th century BCE	*Greece*	*Aristotle*	*"Natural place" theory — early attempt to explain falling.*
3rd century BCE	*Greece*	*Archimedes*	*Buoyancy, hydrostatics — groundwork for gravitational understanding.*
2nd century CE	*Greece*	*Ptolemy*	*Geocentric orbital model with mathematical sophistication.*
9th–12th century	*Islamic Golden Age*	*Al-Kindi, Al-Farabi*	*Writings on tendencies and attraction of bodies.*
11th century	*Persia*	*Ibn Sina*	*Impetus theory — precursor to inertia.*
11th century	*Persia*	*Al-Biruni*	*Compared Indian & Greek cosmologies; noted Earth's gravity.*
14th century	*Syria*	*Ibn al-Shatir*	*Advanced planetary models (pre-Copernican).*

14th century	*Europe*	*Nicole Oresme*	*Early ideas on planetary motion and impetus.*
Late 16th century	*Denmark*	*Tycho Brahe*	*Precise planetary data → foundation for Kepler.*
1543	*Poland*	*Nicolaus Copernicus*	*Heliocentric model of solar system.*
Early 17th century	*Germany*	*Johannes Kepler*	*Laws of planetary motion (elliptical orbits).*
1609–1610	*Italy*	*Galileo Galilei*	*Acceleration of free fall; inertia; observation of celestial motion.*
17th century	*England*	*Robert Hooke*	*Proposed inverse-square attraction before Newton.*
1687	*England*	*Isaac Newton*	*Philosophiæ Naturalis Principia Mathematica — universal law of gravitation.*
18th century	*England*	*John Michell*	*Predicted black holes ("dark stars") with Newtonian gravity.*
18th century	*France*	*Émilie du Châtelet*	*Expanded Newton's work; energy and motion insights.*

Gravity - Activities